

Oliver Dippel

Research Engineer · Reinforcement Learning & Foundation Models

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PROFILE

PhD-trained research engineer working on reward design, failure-mode diagnosis, and generalization in RL and foundation models, with a peer-reviewed publication record (AAMAS 2026, SGAI 2024/2023). Combines statistical rigor with hands-on distributed training infrastructure (JAX/PyTorch, Ray, HPC/SLURM) and has translated research directly into deployed decision-support and agentic systems.

EXPERIENCE

Research Engineer — RL & Foundation Models (PhD)

University of Liverpool / CDT Distributed Algorithms, Industry-Funded · Dec 2021 – 2025

- Designed the Heuristic Transformer, a Bayesian RL architecture, achieving improved zero-shot policy generalization over baseline RL methods in peer-reviewed benchmark evaluations (AAMAS 2026).
- Resolved recurring training-time anomalies — reward hacking, posterior collapse, entropy collapse — through systematic ablation studies and statistical root-cause diagnosis rather than hyperparameter patching.
- Deployed a model fine-tuned on structured expert feedback as a production decision-support tool, sustained in operational use among domain specialists, by translating expert feedback loops into a real-time recommender system.
- Cut experiment turnaround time substantially across HPC job cycles by architecting fault-tolerant, reproducible distributed training and MLOps pipelines in JAX and PyTorch with Ray.

Research Engineer — LLM Agents & Agentic Systems

ZeroAI · Liverpool, UK (part-time, concurrent with PhD) · Sep 2023 – Dec 2024

- Designed a drift-detection and regression-testing harness for agent behavior under distribution shift, detecting distinct agent failure modes ahead of deployment via a systematic adversarial evaluation suite.
- Integrated diffusion models into a production perception pipeline, strengthening multi-step trajectory forecasting in high-dimensional sensor state-spaces and improving forecast stability under distribution shift.
- Engineered tool-use and structured-output validation on AWS infrastructure for a safety-critical autonomous-driving agent system, with CI/CD-integrated monitoring and observability.

Independent Technical Advisor — Agentic Systems (Part-time)

Curate IQ · Seattle, WA, USA (Remote) · Jun 2026 – Present

- Designed the intent-routing and retrieval-augmented generation architecture for a B2B agentic platform, fusing catalog, safety, and industry-knowledge sources for correct multi-skill hand-off.
- Built a RAG pipeline sourced directly from live product data, keeping every generated answer strictly grounded in the merchant catalog.

Data Science Intern — Applied ML

Novartis Pharma AG · Basel, Switzerland · Jun 2020 – Dec 2020

- Validated ML pipelines against GxP data-quality standards in a regulated healthcare environment.

Statistical Consulting & ML Teaching

Freelancer / statistiknachhilfe.ch · Jan 2021 – Present

- Taught Bayesian and probabilistic ML methods to technical and non-technical audiences; designed the underlying curriculum.

SKILLS

RL & Training: Policy gradient methods, RLHF, reward design, credit assignment, entropy regularisation, exploration strategies, Bayesian RL

Time-Series & Failure Diagnosis: Distribution-shift detection, anomaly and drift diagnosis, reward hacking, posterior collapse, entropy collapse, gradient instability — root-cause analysis

Languages & Stack: Python (advanced), JAX (Haiku/Flax), PyTorch, AWS, Docker, Ray, SLURM/HPC, Weights & Biases, Git/CI-CD

Agentic Systems: Tool-use agents, multi-step reasoning, retrieval-augmented generation, structured output, multi-agent orchestration, evaluation pipelines

Production AI: Regulated ML deployment (Novartis), safety-critical production AI (ZeroAI), B2B plug-and-play GenAI (Curate IQ)

PUBLICATIONS

AAMAS 2026 — Heuristic Transformer: a Bayesian RL architecture for zero-shot policy generalization. SGAI 2024, SGAI 2023 (details on request / GitHub).

EDUCATION

Ph.D. Electrical & Control Engineering — University of Liverpool / CDT Distributed Algorithms (2021–2025), Industry-Funded

M.Sc. Applied Statistics — University of Göttingen (2017–2020)

B.A. Economics — University of Göttingen (2014–2017)

LANGUAGES

German (native) · English (fluent) · French (intermediate)